

Treat the following exercises:

Exercise: 1 (6 pts)

- Write the corresponding DTD file for the following xml document. (It must use ENTITY)
- In the XML document add the necessary tag which insures the link of the XML file to your DTD file.

```
<?xml version="1.0" standalone="yes" ?>
<COLLECTION>
<HOUSEPLANT>
  <COMMON> Bloodroot </COMMON>
  <BOTANICAL> Sanguinary Canadensis </BOTANICAL>
  <ZONE NUM="4"/>
  <LIGHT>Mostly Shady</LIGHT>
  <PRICE currency="€" >2.44</PRICE>
  <AVAILABILITY>599</AVAILABILITY>
</HOUSEPLANT>
<PLANT>
  <COMMON>Columbine</COMMON>
  <BOTANICAL>Aquilegia Canadensis</BOTANICAL>
  <ZONE NUM="3"/>
  <LIGHT>Mostly Shady</LIGHT>
  <PRICE currency="$" >9.37</PRICE>
  <AVAILABILITY>699</AVAILABILITY>
</PLANT>
<HOUSEPLANT>
  <COMMON>Marsh Marigold</COMMON>
  <BOTANICAL>Caltha palustris</BOTANICAL>
  <ZONE NUM="4"/>
  <LIGHT>Mostly Sunny</LIGHT>
  <PRICE currency="$" >6.81</PRICE>
  <AVAILABILITY>799</AVAILABILITY>
</HOUSEPLANT>
</COLLECTION>
```

Exercise: 2 (6 pts)

- Write the corresponding schema file for the following xml document.
- In the XML document add the necessary attributes to the root element which insures the link of the XML file to your schema file.

```
<?xml version="1.0" standalone="yes" ?>
<breakfast_menu>
<diet>
  <name>Belgian Waffles</name>
  <price currency="L.L." >19000</price>
  <description>
    Two of our famous Belgian Waffles with plenty of real maple syrup
  </description>
  <calories value="650"/>
</diet>
<nutrition>
  <name>Strawberry Belgian Waffles</name>
  <price currency="L.L." >14000</price>
  <description> Light Belgian waffles covered with strawberries and whipped cream
  </description>
  <calories value="900"/>
</nutrition>
```

```
<nutation>
  <name>Homestyle Breakfast</name>
  <price currency="L.L.">10000</price>
  <description>
    Two eggs, sausage, toast, and our ever-popular hash browns
  </description>
  <calories value="950"/>
</nutation>

<diet>
  <name>Berry-Berry Belgian Waffles</name>
  <price currency="L.L.">14000</price>
  <description>
    Light Belgian waffles covered with an assortment of fresh berries and whipped cream
  </description>
  <calories value="300"/>
</diet>
</breakfast_menu>
```

Exercise: 3 (3pts)

- Write a CSS file for the following XML instance, to do the following formatting on the contents of tags.
 - The ARTIST element is bold.
 - The TITLE in double border.
 - The color of PRICE is blue.
 - COUNTRY is in *italic*.
 - COUNTRY and COMPANY are underlined.
 - All font size is 20pt.
 - The background color is red.
 - The font family of TITLE, COMPANY and YEAR is Courier New.
- Write the corresponding tag in the XML instance to link it to the ccs file:

```
<?xml version="1.0" standalone="yes" ?>
<CDS>
  <CD>
    <TITLE>Greatest Hits</TITLE>
    <ARTIST>Dolly Parton</ARTIST>
    <COUNTRY>USA</COUNTRY>
    <COMPANY>RCA</COMPANY>
    <PRICE>9.90</PRICE>
    <YEAR>1982</YEAR>
  </CD>
  <CD>
    <TITLE>Eros</TITLE>
    <ARTIST>Eros Ramazzotti</ARTIST>
    <COUNTRY>EU</COUNTRY>
    <COMPANY>BMG</COMPANY>
    <PRICE>9.90</PRICE>
    <YEAR>1997</YEAR>
  </CD>
</CDS>
```

Exercise: 4 (5pts)

Apply the following XSL file on the XML file of exercise 3. Then draw and write the web site result of it:

```
<?xml version="1.0" ?>
<?xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform" version="1.0" ?>
<xsl:output method="html" />
<xsl:variable name="label"> XSLT: Table </xsl:variable>
<xsl:template match="/">
<html>
<head>
<title> <xsl:value-of select="$label"/></title>
</head>
<body>
<h1><xsl:value-of select="$label" /> </h1>
<table cellpadding="5" border="1">
<tr>
<th> TITLE </th>
<th> ARTIST </th>
<th> YEAR </th>
<th> PRICE </th>
</tr>
<xsl:apply-templates select="//CD"/>
</table>
</body>
</html>
</xsl:template>
<xsl:template match="CD">
<tr bgcolor="#cccccc" >
<td> <i> <xsl:value-of select="TITLE" /></i></td>
<td> <xsl:value-of select=" ARTIST " /> </td>
<td> <xsl:value-of select=" YEAR " /> </td>
<td> <xsl:value-of select=" PRICE " /> </td>
</xsl:template>
</xsl:stylesheet>
```